A clearer picture of change.

PROJECT OVERVIEW / EVALUATOR EDITION

PROHOR is an offline-capable health-monitoring prototype that helps a crew member organize personal indicators, compare them with their own prior observations, and prepare context for a human reviewer.

At a glance. A deterministic personal-baseline engine, a navigable health atlas, a local logbook, NASA research context and a structured handover — designed for human interpretation rather than automated diagnosis.

THE CHALLENGE

On long missions, crew need a way to notice meaningful changes when immediate specialist support may be limited. PROHOR turns a set of measurements into an understandable, traceable workflow: observe, compare, inspect context and share for review.

TRY IT IN THREE STEPS

01 Observe. Open Mission overview and choose "See the change" in the guided tour. The synthetic scenario selects CDR-01, flight day 146; inspect the sleep change in Trends.

02 Understand. Review the prior-day median, coverage and rule explanation. Visit NASA evidence to explore the historical OSD-575 metabolic panel separately from the demo.

03 Prepare. Open Crew handover to inspect and export a structured summary. Data & settings provides local CSV import/export and controls.

Open PROHOR · Source repository · Official challenge

Where the data comes from.

INTEGRATIONS / TRACEABILITY / BOUNDARIES

NASA OSD-575 | analyzed historical data

The evidence explorer parses NASA's original Inspiration4 submitted comprehensive metabolic panel. It contains 531 rows, 19 analytes and four coded subjects sampled before and after a three-day 2021 mission. It is historical research, not live crew telemetry or validation for a Mars mission.

Original file: LSDS-8_Comprehensive_Metabolic_Panel_CMP.upload_SUBMITTED.csv
SHA-256 (25,275 bytes):

dab247eb794edab59aaaa683c85fd1f2dbf4ae3684872739338e10ead4192a0c

NASA study and source · Original CSV download · Provenance documentation

Related NASA studies | metadata only

OSD-569 and OSD-571 provide titles, descriptions and archive links in a curated library; they do not supply personal readings or additional chart values. OSD-87 is optional mouse-study metadata. NASA Human Research Program materials supply hazard and sleep context, not approval of PROHOR's thresholds.

NOAA space weather | Earth context

The NOAA SWPC planetary Kp feed provides Earth geomagnetic context, with source and observation times. It is not Mars or cabin conditions, a crew radiation dose, or evidence that space weather caused a health change.

Synthetic and local | kept separate

The built-in mission is a deterministic synthetic scenario, not real astronaut health data. Users may enter or import their own readings locally. NASA research records do not enter the personal-baseline engine.

Honest status. Network responses and dated fallback snapshots are labeled. If unavailable, feeds fall back to cached data rather than inventing observations; the source observation may be older than retrieval.

How the system works.

METHOD / PRIVACY / RESILIENCE

From observation to review

Local readings → validation and unit conversion → prior-day baseline → directional change band → plain-language trend and handover. Data stay in the browser workspace; there is no health-record submission endpoint or remote health synchronization.

The baseline rule

For local inputs, use flight days d−30 through d−1, with the latest valid observation per day and at least seven distinct prior days. Compute the median and interquartile range (IQR). The median-centered band has width max(1.5 × IQR, a signal-specific noise floor). A value outside its configured direction is flagged for review.

The synthetic replay uses a fixed enrollment window, FD 111–140, for scenario days 142–147. Seven to 20 days are labeled provisional; 21–30 established. Those labels and cutoffs are developer choices, not clinical confidence or safety thresholds.

NASA explorer method

The OSD-575 browser calculates each chosen subject's pre-flight median and descriptive change from that median. The CSV checksum is checked where Web Crypto is available; a mismatch blocks analysis. Limited samples, lab intervals and missing values must be interpreted in their original context.

Architecture and availability

React, TypeScript and Vite run the interface and analysis. Same-origin serverless /api/feeds adapters retrieve NASA OSDR study metadata and NOAA Kp. A service worker precaches the app, data and this brief after an initial successful online visit; browser storage restrictions or cache eviction can prevent offline use.

Privacy and caution. Browser storage is device-local but not a secure medical record: shared devices, browser extensions, exports and backups can expose data. Do not enter sensitive real health information without an appropriate security review.

Read the full methodology and signal-specific rules

Evidence and next steps.

VALIDATION / LIMITATIONS / REFERENCES

What the prototype demonstrates

A reproducible synthetic walk-through, separate local data entry and CSV validation, NASA source exploration, a Kp context feed, and a structured handover. The repository contains unit and browser tests for baseline behavior, import, source parsing, navigation and offline workflows.

What it does not establish

This is not a diagnostic device, prediction model, treatment recommendation or fitness-for-duty assessment. The change rules have not been clinically validated. A stable baseline can still be unhealthy; missing values are not normal values. Human judgment and applicable mission medical procedures take precedence, especially for sudden or severe symptoms.

Responsible next steps

Clinical and human-factors evaluation; threat modeling and protected data handling; sensor integration only after engineering and privacy review; prospective validation with appropriate permissions. These are future work, not current features.

PRIMARY REFERENCES

- NASA OSD-575 — Inspiration4 serum metabolic panel
- NASA OSD-569 — related whole-blood study · OSD-571 — human plasma study
- NOAA SWPC — planetary K index · NASA HRP — human spaceflight hazards
- Source, tests and documentation

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AI transparency: AI-assisted tools were used in website updates, documentation and presentation production. The team should review and complete its own project-submission disclosure, including all tools actually used.